



SECP2523: DATABASE

SEM I 2024/2025

DATABASE DESIGN DESCRIPTIONS: CONCEPTUAL DATABASE DESIGN FOR KADA'S COOPERATIVE SYSTEM

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NOTE: A module can consist of one or multiple (related) use case. A team member in a group must be responsible to at least one module.

1. Introduction

1.1 *Overview about the company*

Kemubu Agricultural Development Authority (KADA) is a prominent agricultural development organization in Malaysia, established to support and develop the agricultural sector in the Kelantan region. Operating under the Ministry of Agriculture and Food Industries, KADA plays a crucial role in implementing agricultural development programs, managing irrigation systems, and supporting local farmers through cooperative initiatives.

The organization's cooperative division manages membership applications, coordinates agricultural support programs, and facilitates member services. Currently serving thousands of farmers across the Kelantan region, KADA's cooperative system is integral to providing financial assistance, technical support, and agricultural resources to its members.

1.2 *Problems*

a. Background of existing system

The cooperation system used by KADA is currently being done manually. All of the forms and paperwork are submitted and stored in physical by the client and the organization. Applicants need to attend the KADA office to obtain the application form, and there is no computerized system for the submission of the documents required. The application type is divided into two, member application and loan application. If a client wants to apply for a loan, one must first become a member of KADA cooperation. The applicant needs to submit a member application and pay RM50 to become a member. When the application for membership is approved, then only they can proceed with loan application. Applicant's information is kept in a physical paper and the clerk must check their application one by one. Then, board members of KADA will conduct a meeting weekly to verify the application and decide whether to accept or reject the application. Once the result is finalized, the applicant will be notified through SMS messages. KADA offers a various type of loan for its member including Al-Bai, Al-Innah, Vehicle Restoration, Road Tax & Insurance and Al-Qadrul Hassan. Applicants can choose what type of loan they want to apply for according to their needs. The processes for loan application are quite similar to member application where applicant must come to the KADA office and obtain the application form, thus submit it with all the documents related at the office physically. A meeting between board members will be conducted also to determine the decision for the application. Lastly, successful applicants will be notified through SMS messages.

In addition, KADA's cooperation system includes a shares system which all members are eligible to join. Members can contribute to the organization's capital through purchasing of shares. These shares represent their ownership in the organization. Upon first time to apply for membership, the applicant needs to pay RM50. The registration fee will then be recorded as the applicant's shares when they are successful in becoming a member. Additionally, the shares amount will also put into factors which the member is eligible to apply for loan or not, all members must have at least RM350 worth of shares for them to apply for loan. If their shares do not fulfill the condition, they must first purchase the shares until they reach the required minimum amount.

b. Issues with existing system

The issues arising from the manual handling of the application which are:

- i) **Data Security & Reliability** – All the related documents containing sensitive information about customers were stored physically. There might be risk of losing or leaking the data to malicious party because of human error. Moreover, the organization of the documents might not be in a systematic order, and it is possible for the data to mistakenly match with the wrong customer and end up providing misinformation about the customer. Thus, understanding the importance of having a convenient system in an organization might help the organization to provide secure and more reliable data for the customer.
- ii) **Timeliness** – The process of requiring and submitting documents is time consuming as the applicant needs to travel to KADA office multiple times. It requires a large amount of time just for the enquiries and submission of the documents. Some of the members might be living far away across the country and the distance between KADA office and their hometown must be put into consideration. To put it simply, the organization must provide a centralized system for the documentation which is accessible by the customer whenever or wherever they want.
- iii) **Scalability** – As time grows, KADA might experience an increasing number of its members. The manual system that is currently being used may struggle to handle a large volume of requests and applications in the future. Besides, the weekly meetings for applications approval might become a bottleneck and lead to slower response to the applicant, which will frustrate applicants. In summary, KADA needs to computerize their systems to give the best customer experience and enhance the efficiency of the system.

1.3 Proposal

The proposed cooperative system for KADA is a digital portal designed to simplify and streamline membership and loan applications processes. Applicants can apply for membership online, and once approved, they become members and gain access to loan services. Admins manage and verify applications, while Board Members (ALK) are responsible for making final member and loan approval decisions. The system will help to improve efficiency, provides secure data storage, and enhances the overall user experience.

The system will serve four types of users:

1. Applicants:

- Individuals applying for KADA Cooperation membership
- Applicants must submit an online application form and pay the required membership fee
- They receive real-time updates about their application status

2. Members:

- Once approved, applicants become members and can apply for loans anytime
- Members can use the loan calculator to estimate repayment plans and track their loan application statuses.

3. Admins:

- Admins handle all administrative tasks, including verifying membership and loan applications.
- They update applications statuses and forward approved loan applications to the Board Members (ALK) for final decisions.
- Admins also manage user data and ensure smooth operation of the system.

4. Board Members (ALK):

- Board Members review loan applications forwarded by Admins.
- They make the final decision on loan approvals or rejection based on eligibility criteria and KADA policies.

Key Features:

- Online membership registration for applicants.
- Loan application and tracking for members.
- Real-time updates on application status via email notifications.
- A loan calculator to help members plan repayment schedules.
- A centralized system for admins and board members to manage and review applications.

1.4 Definitions, Acronyms and Abbreviations

2. Database System Requirements

2.1 System Overview

The system will be a web-based cooperative system which automates all the processes that happen in this system. The system consists of four modules which are membership application, loan application, loan calculator and shares features. The system will have four entity or type of users which are applicant, members, admin and board members (ALK). Each of the entity have their own roles which described as follows:

- Applicant – apply for membership and track the progress of the application
- Members – apply for loan and track the progress of the application, access to loan calculator features, contribute shares
- Admin – Manage membership and loan applications, generate report, communicate with board members
- Board Members (ALK) – Review membership and loan applications, approve or reject the applications, communicate with admin

2.2 Module <Membership Application>

2.2.1 Module Description

The membership application module is designed to streamline and automate the process of registering new members for KADA's cooperatives. This module allows applicants to fill out an online form, upload necessary documents, and make the fee payment through secure digital methods.

Administrators can manage and validate the application, updating its status as "Reviewed" or "Not Reviewed Yet". Applicants are notified of their application status through email, improving communication. However, administrators are not authorized to approve or reject the applications. This module eliminates the need for manual form submission, reducing processing time and improving data security. The use cases involved in this module are:

UC001: Apply for Membership

- Includes: Fill personal details, upload necessary documents, and make fee payment.
- Description: Applicants submit the required information, which is validated and stored in the database.

UC002: Pay for Membership Fee

- Description: Applicants pay the RM50 membership fee digitally, and the payment is recorded for verification.

UC008: Notify Application Status

- Description: The system sends notifications to applicants via email regarding their application status.

2.2.2 Transaction Requirement: Data Entry

1. Applicants fill out an online membership application form with personal details such as:

- Name
- Address
- Contact Number
- Email
- Identification Card Number

2. Applicants upload supporting documents like proof of identity.

3. Payment of the RM50 membership fee is recorded.

4. The system automatically saves the submitted application data in the database.

2.2.3 Transaction Requirement: Data Update/Deletion

1. Administrators can update applicant details in case of errors or missing information.

2. Administrators can delete incomplete or fraudulent applications.

3. Administrators can update the application status to “Reviewed” or “Not Reviewed Yet”.

4. Applicants may update their personal information before approval (contact number or address)

5. Approved membership records are locked in updates except by administrators.

2.2.4 Transaction Requirement: Data Queries

1. Retrieve all pending membership applications for administrator review.
2. Search for specific applicants based on name, email, or identification card number.
3. Display the status of applicant (“Reviewed” or “Not Reviewed Yet”).
4. Generate a list of all approved members, including their membership IDs.
5. Generate reports on membership application trends (daily, weekly, or monthly submission).

Proposed Data Requirements

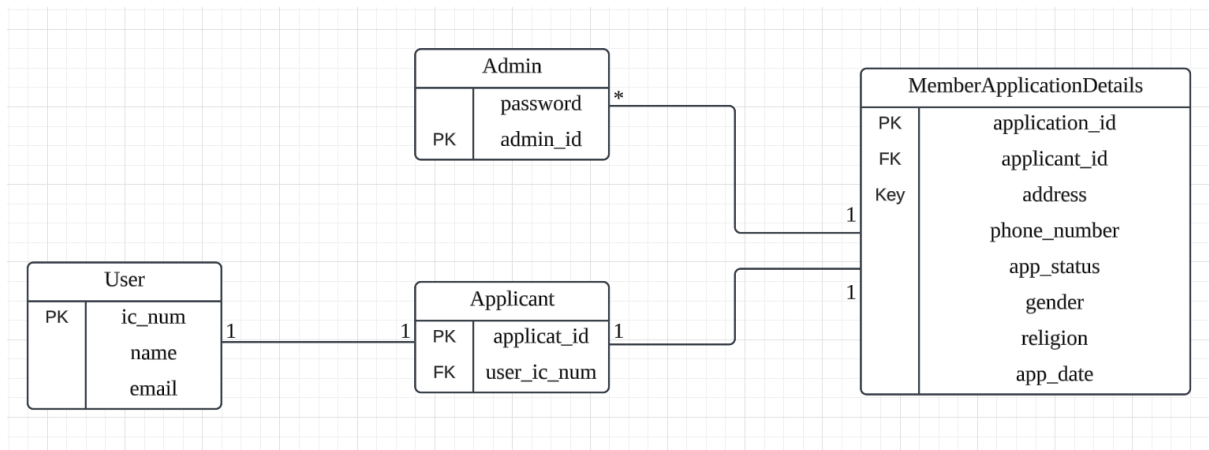
The **User** entity stores basic user information required for membership applications. This includes the User ID as the primary key, which uniquely identifies each user in the system. Other attributes include name, email, and IC number. The name attribute can be split into first name and last name for better organization. A user can apply for one or more membership applications, linking to the Applicant entity.

The **Applicant** entity holds data specific to individuals applying for membership. It includes the Applicant ID as primary key, and a foreign key that links the Applicant entity to the User entity. Each applicant is uniquely identified, and their application data is securely stored. One user can correspond to one applicant, ensuring that an individual’s application information is organized and associated with their user account.

The **MemberApplicationDetails** entity contains all relevant details about the membership applications. These include the Application ID as primary key, address, number phone, gender, religion, application status, and application date. A foreign key links this entity to the Applicant entity, ensuring that each application is tied to a specific individual. One applicant can submit one membership application at a time, which is tracked by this entity. Application statuses such as “Reviewed” or “Not Reviewed Yet” are also recorded here.

The **Admin** entity stores information about administrators responsible for validating membership applications. It includes the Admin ID as primary key, along with attributes such as name and password. An admin can validate multiple membership applications, but their role is limited to marking application as “Reviewed” or “Not Reviewed Yet”. They cannot approve or reject applications.

2.2.5 Local Conceptual Data Model (ERD)



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2.3 Module <Loan Application>

2.3.1 Module Description

The loan application module is designed to facilitate and manage loan requests efficiently. This module allows members to apply for loans, provide detailed loan information, and track the status of their applications. Administrators can review loan applications and update their status but are not authorized to approve or reject them. The loan status will be determined by the board members who are in charge of managing reviewed application. Applicants are notified about the status of their applications through automated notifications. This module ensures a smooth and secure process for loan management.

UC015: Apply for Loan

- **Includes:** Submit loan application and provide necessary member information.
- **Description:** Members submit loan applications, which are validated and stored in the database.

UC016: Fill in Loan Details

- **Description:** Members specify the loan amount, repayment period, purpose, and other details.

UC008: Notify Application Status

- **Description:** The system sends notifications to members via email about their application status.

UC018: Loan Repayment

- **Description:** Members repay loans in installments, and repayments are recorded in the system

2.3.2 Transaction Requirement: Data Entry

1. Cooperative member will submit loan application form with such details:

- Member ID
- Loan Amount
- Loan Duration
- Bank statement
- Payslip
- Personal Information
- Guarantor Information

2. Members will fill in the loan details, including the requested loan amount and purpose of loan
3. The system automatically saving the submitted loan applications data in the database

2.3.3 Transaction Requirement: Data Update/Deletion

1. Admin can update the loan details in case of any errors of information from the applicant side.
2. Admin can delete or order to resubmit incomplete or fraudulent information of applications.
3. Admin will update pending applications status to “Reviewed” or “Not Reviewed Yet”.
4. Members may update their loan applications details before approval in case of any errors.
5. Approved loan records are locked in updates except by administrators.

2.3.4 Transaction Requirement: Data Queries

1. Board members retrieved all reviewed loan applications from admin.
2. Search for specific loan applications based on Member ID or application date.
3. Display the status of loan application either “approved” or “rejected”.
4. Generate a list of approved loans with their loan applications details.
5. Generate reports on loan application trends and an overall summary for members of cooperative to view.

Proposed Data Requirements

The **Admin** entity is responsible for managing and overseeing loan applications. It includes the admin_id as the primary key, which uniquely identifies each admin, and a password attribute to securely store their credentials. Admins have the role of managing multiple loan applications, primarily focusing on reviewing and updating the application statuses as needed.

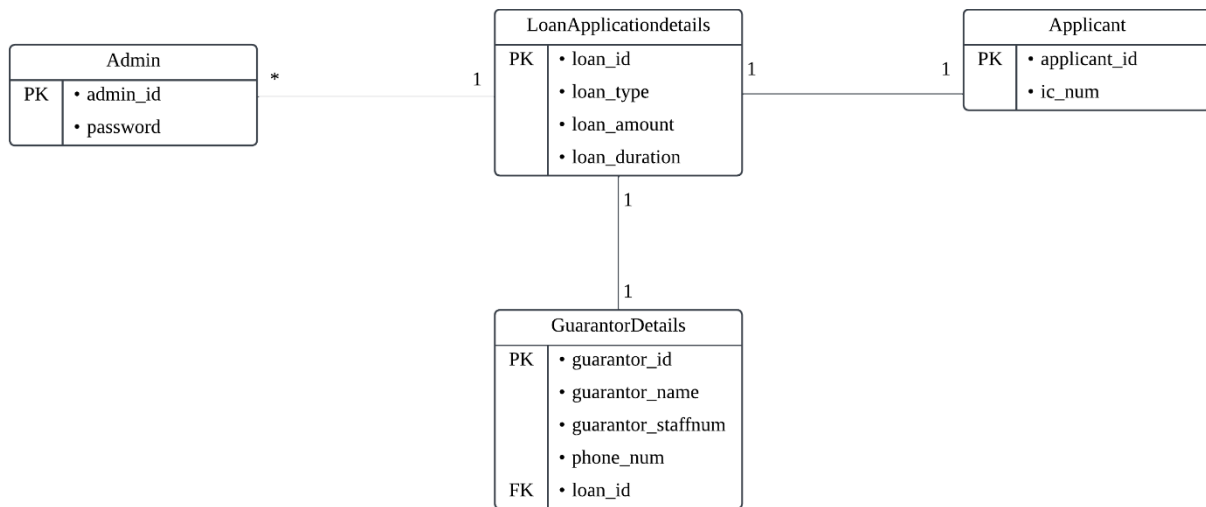
The **Applicant** entity stores data specific to individuals applying for loans. This includes the applicant_id as the primary key, which uniquely identifies each applicant, and the ic_num, which stores the applicant’s national identification card number. Each applicant corresponds to exactly one loan application, ensuring that all their details are clearly associated with their specific application.

The **LoanApplicationDetails** entity holds detailed information about loan applications. This includes the loan_id as the primary key, which uniquely identifies each loan application, along with attributes such as loan_type (indicating the type of loan, e.g., personal or housing), loan_amount (the amount requested), and loan_duration (the repayment period). This entity is linked to both the Applicant and Admin entities, ensuring that each loan application is associated with a specific applicant and can be reviewed by an admin.

The **GuarantorDetails** entity records information about the guarantor associated with each loan application. It includes the `guarantor_id` as the primary key, which uniquely identifies each guarantor, along with attributes such as `guarantor_name` (the guarantor's full name), `guarantor_staffnum` (their unique staff number, if applicable), `phone_num` (their contact number), and a foreign key, `loan_id`, linking the guarantor to a specific loan application. Each loan application requires a guarantor, forming a one-to-one relationship between the loan application and the guarantor.

The relationships among these entities are structured as follows: One admin can manage multiple loan applications, but each loan application is reviewed by a single admin. One applicant corresponds to one loan application, establishing a one-to-one relationship between the Applicant and LoanApplicationDetails entities. Similarly, each loan application must have one guarantor, creating a one-to-one relationship between LoanApplicationDetails and GuarantorDetails. This structure ensures a logical and organized flow of data while maintaining the integrity of loan application records.

2.3.5 Local Conceptual Data Model (ERD)



2.4 Module <Loan Calculator>

2.4.1 Module Description

Loan Calculator module allows users to calculate their estimated loan repayments by filling in the details required in the calculator. The details include desired loan type, desired loan amount, loan interest and loan duration. This module helps users to finalize their decisions before applying for loan and visualize the amount that they need to repay back after successfully getting the loan.

2.4.2 Transaction Requirement: Data Entry

Members input the following details into the calculator form:

- Loan type – Type of loan that members wish to apply
- Loan amount – Total amount of desired loan that members want to borrow
- Loan period – Duration of the loan
- Loan interest – Rate of interest of the loan (fixed from the organisation)

The data that members input above will be stored in the *loanCalc_history* table for record and history purposes. This will allow members to see their past calculations history. Additionally, the data will be stored into *loan_calculation* table temporarily, to allow the system to retrieve the previous inputs when members revisit the loan calculator page.

2.4.3 Transaction Requirement: Data Update/Deletion

Update:

- Member can modify the details before the calculator generate an estimation
- Members can adjust or modify the loan amount if they want to test different scenarios
- Members can change the loan period and loan interest to see the how it affects the estimation

Delete:

- Provide an option for members to reset the calculator forms, clear all input fields
- Automatically clear the calculation data if members log out

All the modifications will occur on *loan_calculation* tables, as the table stored all the inputs from the members. Thus, the value of each attribute will be updated if there are any changes from the inputs. The table will be cleared once the members log out from the system.

2.4.4 Transaction Requirement: Data Queries

Retrieval:

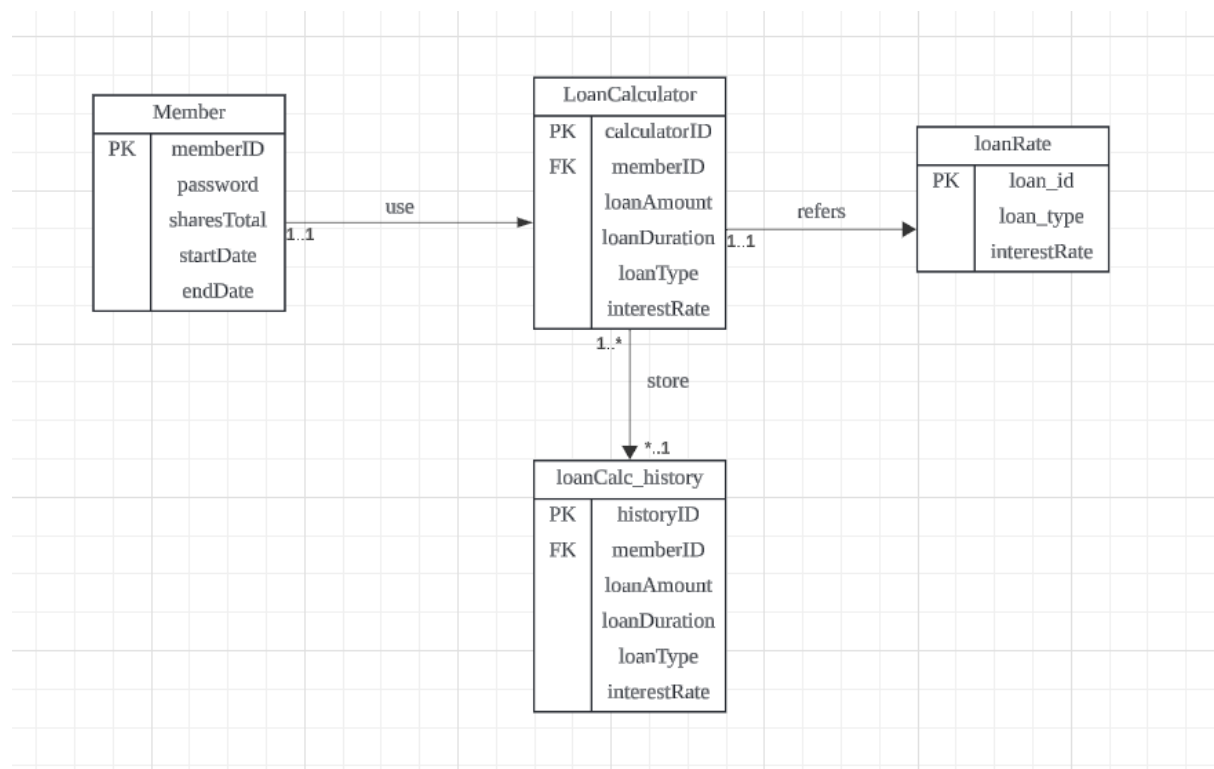
- Retrieve list of interest rates for different type of loan
- If the members navigate back or revisit the loan calculator page, system will retrieve their last entered input for convenience.

Display results:

- Show the calculated monthly repayment and total monthly repayment using a formula
- Generate the loan estimation details to the members

The system will retrieve the interest rate data from *loanRate* table and the data input such as loan amount, loan duration to generate the result and details to the members.

2.4.5 Local Conceptual Data Model (ERD)



2.5 Module <Review Applications>

2.5.1 Module Description

Review Applications module provides Admin and ALK the list of membership and loan applications to be reviewed. This module allows admin to review the applications and validate their details before passing it to ALK. Then, ALK will obtain the list of applications that have been reviewed by admin to either accept or reject. Hence, this will ensure the application process is reviewed and decided in an efficient manner. The use cases that involve in this module are:

UC004: Validate the application

- **Description:** Admin will have the list of member and loan applications that have not been reviewed yet

UC005: Check Application Details

- **Description:** There is a button “view details” for admin to check the application details input by the applicants

UC006: Set the status of application

- **Description:** Admin can set the status of the application to “reviewed” after complete validating and review the application

UC007: Pass the application to ALK

- **Description:** The reviewed applications will be passed to ALK for next process

UC009: Application approval (Accept/Reject)

- **Description:** ALK will have a button to either accept or reject the applications

UC010: Check Application Details

- **Description:** There is a button “view details” for ALK to check the application details input by the applicants

UC011: Pass application status to admin

- **Description:** The status of the application will be passed to admin once ALK accept or reject the application

2.5.2 Transaction Requirement: Data Entry

Admin:

- Admin review the applications and validate details submitted by applicants for membership or loan applications
- Admin will set the **review_status** in *MemberApplicationDetails* and *LoanApplicationDetails* table to either “reviewed” or “not_reviewed”

ALK:

- ALK will check the applications for membership and loan passed by the admin and decide to accept or reject the application
- ALK will set the **status** in *MemberApplicationDetails* and *LoanApplicationDetails* table to either “accepted” or “rejected”

2.5.3 Transaction Requirement: Data Update/Deletion

Admin can update the status field in *MemberApplicationDetails* and *LoanApplicationDetails* table to:

- "Reviewed" after validation
- “Not_reviewed” before validation

ALK can update the **review_status** in *MemberApplicationDetails* and *LoanApplicationDetails* table to either:

- "Accepted"
- "Rejected"

2.5.4 Transaction Requirement: Data Queries

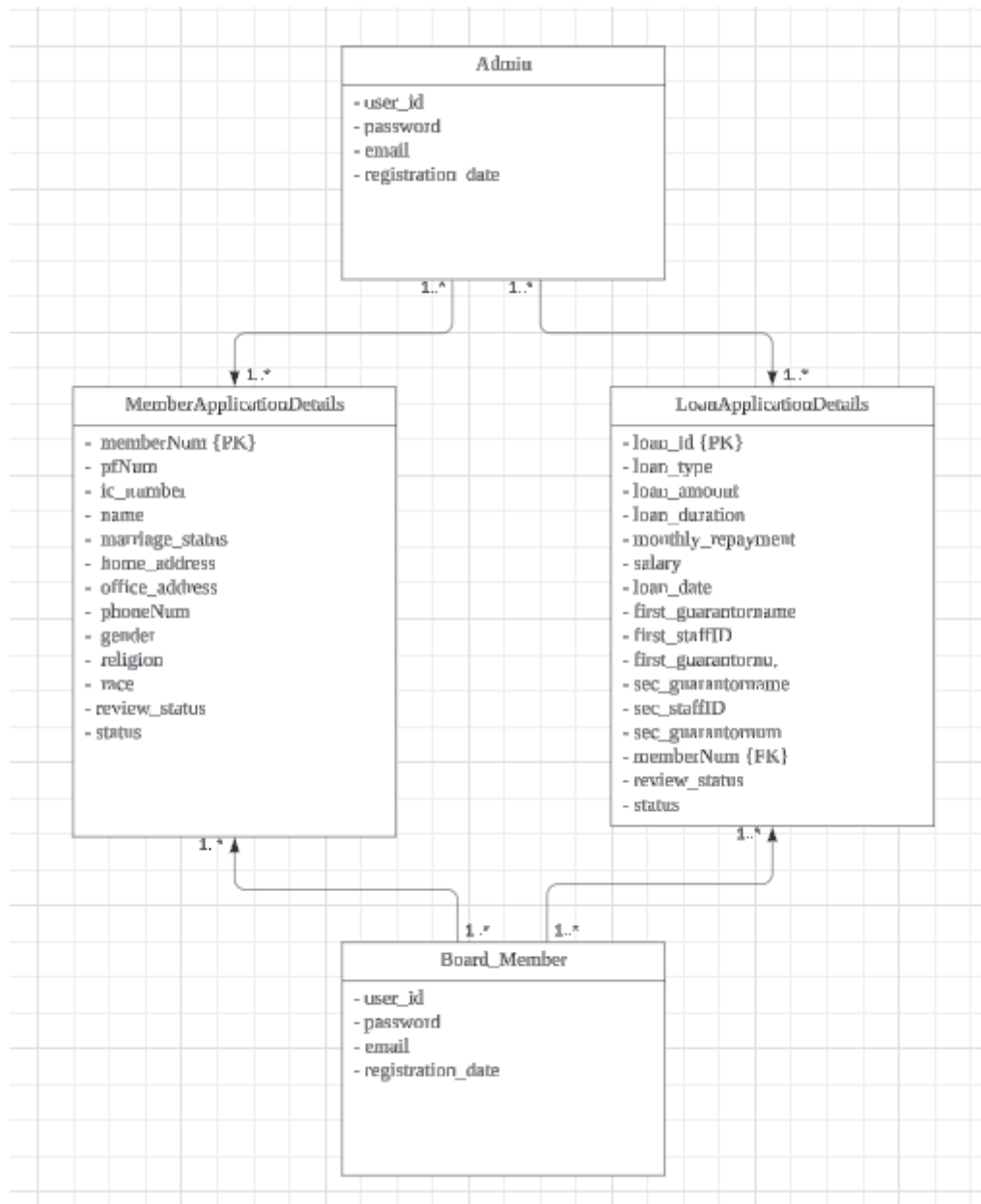
Retrieve Pending Applications:

- ALK can query all applications with status = "pending" and review_status = “Reviewed” from *MemberApplicationDetails* and *LoanApplicationDetails* table.

Retrieve “Not reviewed” Applications:

- Admin can query applications with review_status = "Not_reviewed" from *MemberApplicationDetails* and *LoanApplicationDetails* table.

2.5.5 Local Conceptual Data Model (ERD)



2.6 Module <Saving Applications>

2.6.1 Module Description

The Saving Application module is designed to enable members to deposit funds into their saving accounts conveniently. This module ensures members can apply for savings deposits, provide transaction details, and track deposit statuses. Administrators review deposit applications and validate them but do not have the authority to approve or reject deposits. Members are notified about the status of their savings deposits via automated notifications. The module aims to provide a secure, seamless, and efficient method for managing savings applications.

Use Cases in This Module:

UC022: Apply for Savings Deposit

- **Includes:** Submit savings deposit application and provide necessary member information.
- **Description:** Members submit applications for depositing funds into their saving accounts, which are validated and stored in the database.

UC023: Specify Deposit Details

- **Description:** Members enter the deposit amount, payment method, and other required details.

UC024: Check Deposit Status

- **Description:** Administrators review and validate deposit applications, updating the status to "Accept" or "Reject"

UC025: Notify Deposit Status

- **Description:** Members receive email notifications regarding the review status of their savings deposit applications.

2.6.2 Transaction Requirement: Data Entry

1. Members fill out a savings deposit form with details including:
 - a. Modah Syer
 - b. Modah Fee
 - c. Fixed Savings
 - d. Member Savings
 - e. Welfare Fund Donation
 - f. Payment Method
 - g. Bank Transaction ID (if applicable)
2. The system automatically stores the submitted deposit data in the database after the transaction was verified by admin.
3. The deposited money will appear in the dashboard of said member

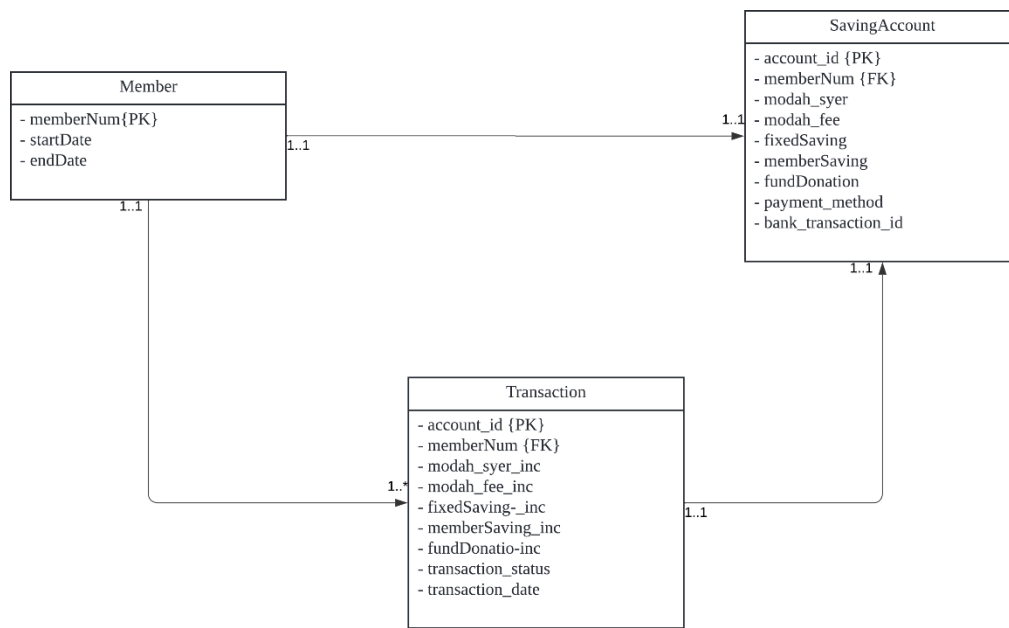
2.6.3 Transaction Requirement: Data Update/Deletion

1. **Administrators:**
 - a. Can update deposit details in case of errors.
 - b. Can delete or request resubmission of incomplete or fraudulent deposit applications.
 - c. Can update pending deposit statuses to "Reviewed" or "Not Reviewed Yet."
2. **Members:**
 - a. May update their deposit application details before the review process in case of errors.

2.6.4 Transaction Requirement: Data Queries

- Retrieve all pending deposit applications for administrator review.
- Search for specific deposit applications based on Member ID or submission date.
- Display the status of deposit applications as either "Reviewed" or "Not Reviewed Yet."
- Generate a list of completed deposit transactions, including Member IDs and deposit amounts.
- Generate reports summarizing deposit trends and providing insights for administrative use.

2.6.5 Local Conceptual Data Model (ERD)



2.7 Module <User Authentication>

2.7.1 Module Description

The User Authentication module is designed to handle login and access control for the KADA cooperative system. It ensures that only authorized users, such as members, administrators, and ALK (Board Member), can access specific functionalities. This module verifies user credentials during login, provides access to their respective modules, and enforces security by preventing unauthorized access.

Use cases in this module includes:

UC0026: Login for Members

- Description: Members enter their staff number and password (identification number) to access their accounts. The system validates their credentials and grants or denies access.

UC0027: Login for Admin

- Description: Administrators log in by entering staff number and password to access administrative functionalities.

UC0028: Login for ALK

- Description: ALK personnel log in by entering staff number and password to access their board members functionalities.

2.7.2 Transaction Requirement: Data Entry

1. Users must enter the following details to log in:

- Staff Number: Unique identifier for the user.
- Password: Secure authentication credential.

2. The system validates the provided credentials against the database.

3. Upon successful login, the session data is stored temporarily to manage user activity.

2.7.3 Transaction Requirement: Data Update/Deletion

1. Update:

- Users cannot update their password because the password is based on their identification number.
- Session data is updated to reflect the user's current activity and login status.

2. Deletion:

- Sessions are automatically deleted upon user logout.
- Admins cannot deactivate user account unless the user explicitly requests it.

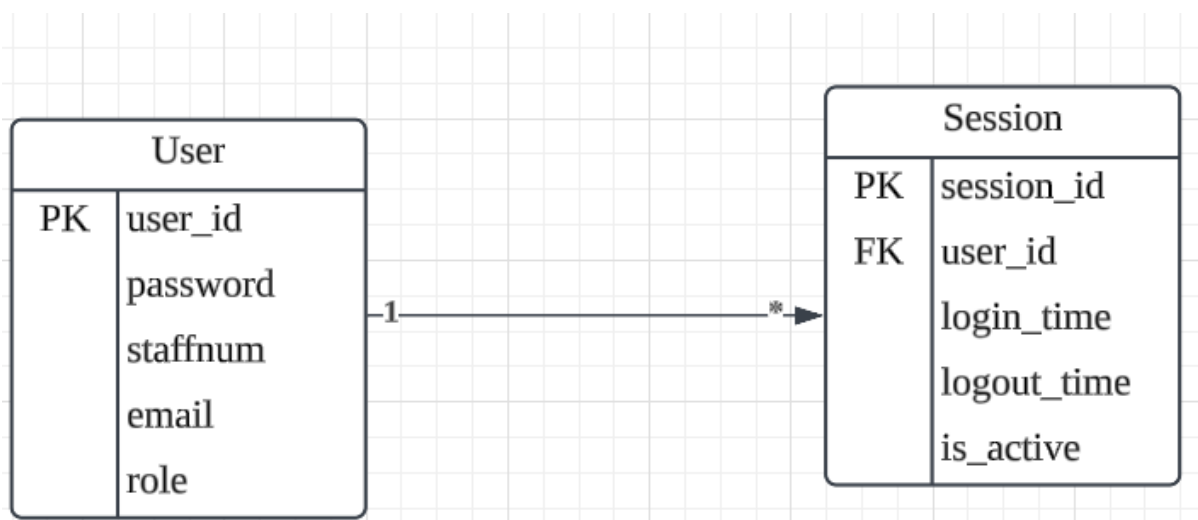
2.7.4 Transaction Requirement: Data Queries

1. **Validate Credentials:** The system validates the staff number and password provided by the user during login by checking against stored records in the database.

2. **Retrieve User Role:** The system retrieves the role of the user (e.g. Member, Admin, ALK) to determine their access permissions.

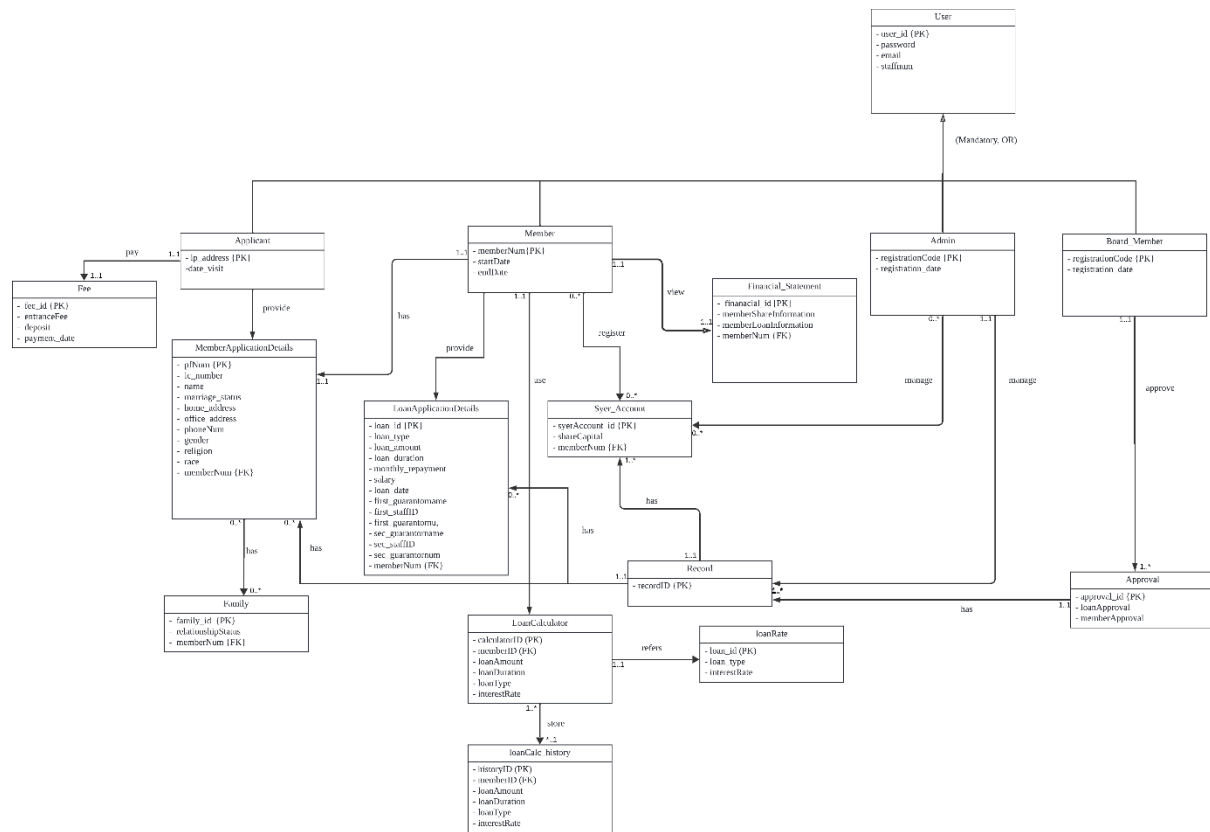
3. **Fetch Active Sessions:** The system can query all currently active sessions, including login times and user IDs, for monitoring purposes.

2.7.5 Local Conceptual Data Model (ERD)

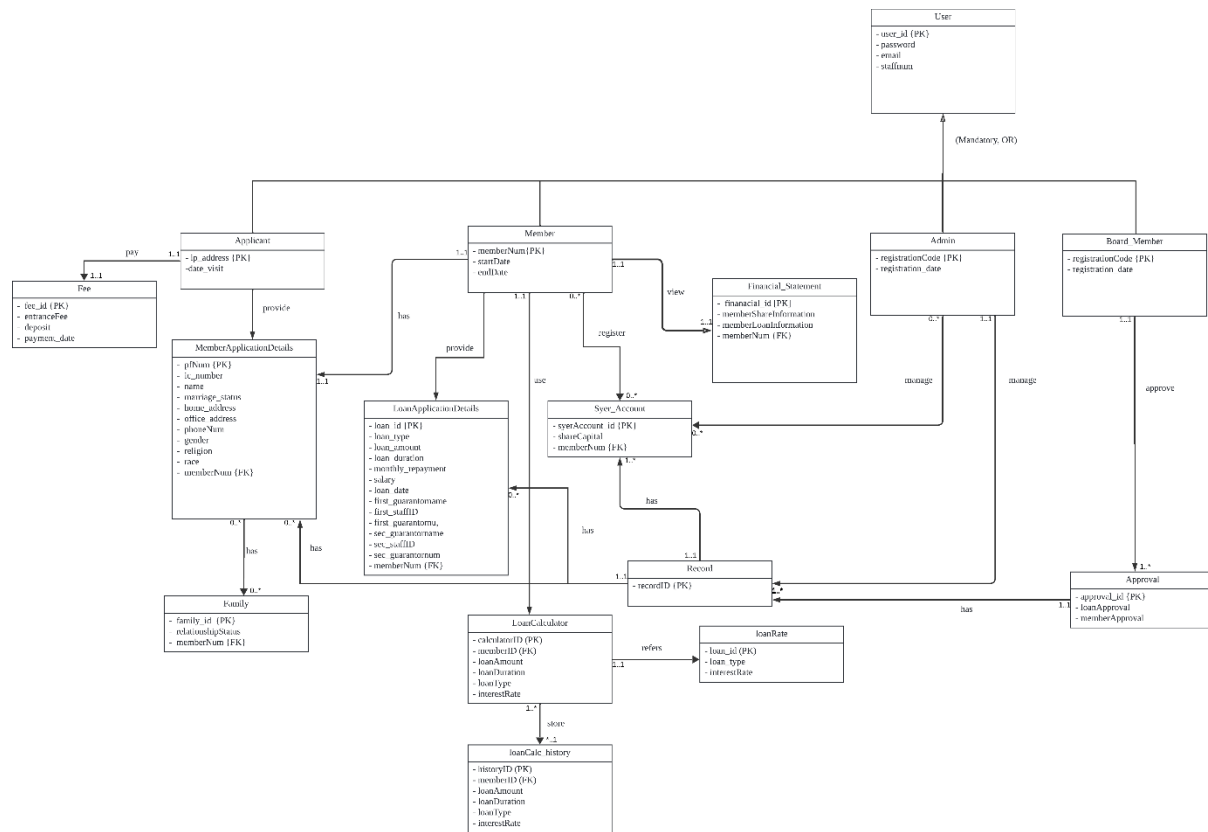


3. Global Conceptual Data Model (Global ERD)

3.1 Global Conceptual Design (ERD)



3.2 Verified Global Conceptual Design (Verified ERD)



3.3 Data Dictionary for the Global Conceptual ERD

1. User

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
user_id	INT	11	Unique identifier for each user. (PK)	1
password	VARCHAR	255	User's password (should be hashed for security).	*hashed value*
email	VARCHAR	255	User's email address.	[email address removed]
staffnum	VARCHAR	20	User's staff number (if applicable, could be a unique identifier for employees).	S1234

2. Member

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
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member Num	VARCHAR	20	Unique identifier for each member.	M20230001
startDate	DATE		Membership start date.	2023-12-01
endDate	DATE		Membership end date.	2024-12-01

3.Admin

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
registrationCode	VARCHAR	20	Unique registration code for each administrator.	ADMIN-0005
registration_date	DATE		Administrator registration date.	2023-09-15

4.Board_Member

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
registrationCode	VARCHAR	20	Unique registration code for each board member.	BOARD-0012
registration_date	DATE		Board member registration date.	2023-08-20

5. Fee

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
fee_id	INT	11	Unique identifier for each fee record. (PK)	1
entranceFee	DECIMAL	11	Amount of the entrance fee.	500
deposit	DECIMAL	11	Amount of the deposit.	200
payment_date	DATE		Date when the fee payment was made.	2023-10-27

6. MemberApplicationDetails

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
pfNum	VARCHAR	20	Unique identifier for each member.	M20230001
name	VARCHAR	255	Member's full name.	Siti Nurhaliza
ic_number	VARCHAR	14	Member's identity card number (Malaysian IC format).	901201-10-5678
marriage_status	VARCHAR	20	Member's marital status (e.g., Single, Married, Divorced).	Married
gender	VARCHAR	10	Member's gender.	Male
religion	VARCHAR	20	Member's religion.	Islam
race	VARCHAR	20	Member's race.	Malay
phoneNum	VARCHAR	20	Member's phone number.	012-3456789
memberNum	INT	11	Foreign key referencing Member.memberNum	1001

7.LoanApplicationDetails

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
loan_id	INT	11	Unique identifier for each loan application. (PK)	501
loan_type	VARCHAR	50	Type of loan (e.g., Al-Bai, Al-Innah, etc.).	Al-Bai

loan_amount	DECIMAL	11	The total amount of the loan.	10000
loan_duration	INT	4	Duration of the loan (in months or years).	24
monthly_repayment	DECIMAL	11	The calculated monthly repayment amount.	458.33
salary	DECIMAL	11	Applicant's salary.	6000
loan_date	DATE		The date the loan application was submitted.	2023-10-27
first_guarantorname	VARCHAR	255	Name of the first guarantor.	Ahmad Ali
first_staffID	VARCHAR	20	Staff ID of the first guarantor (if applicable).	S1234
first_guarantornum	VARCHAR	20	Contact number of the first guarantor.	012-3456789
sec_guarantorname	VARCHAR	255	Name of the second guarantor.	Siti Rahman
sec_staffID	VARCHAR	20	Staff ID of the second guarantor (if applicable).	S5678
sec_guarantornum	VARCHAR	20	Contact number of the second guarantor.	017-9876543
memberNum	VARCHAR	20	Foreign key referencing the Member table, indicating the member associated with this loan application. (FK)	M20230001

8.Syer_Account

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
syerAccountID	VARCHAR	20	Unique identifier for each item account.	001
shareCapital	DECIMAL	11	Share capital amount.	10000
memberNum	VARCHAR	20	Foreign key referencing Member.memberNum.	M20230001

9. Financial_Statement

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
financial_id	INT	11	Unique identifier for each financial statement record. (PK)	12
memberShareInformation	VARCHAR	255	Information about member shares	Share details
memberLoanInformation	VARCHAR	255	Information about member loans	Loan details

10. Family

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
familyID	INT	11	Unique identifier for each family.	201
relationshipStatus	VARCHAR	50	Relationship to the member.	Spouse
memberNum	VARCHAR	20	Foreign key referencing Member.memberNum.	M20230001

11.LoanCalculator

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
calculation_id	INT	11	Unique identifier for each loan calculation. (PK)	301
memberID	VARCHAR	20	Foreign key referencing the Member table, indicating the member associated with this calculation. (FK)	M20230001
loanAmount	DECIMAL	11	The total amount of the loan being calculated.	5000
loanDuration	INT	4	Duration of the loan (in months or years).	24
loanType	VARCHAR	50	Type of loan (e.g., Al-Bai, Al-Innah, etc.).	Al-Innah
interestRate	DECIMAL	5	Interest rate used in the calculation (e.g., 5.25 for 5.25%).	7.5

12.Applicant

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
ip_address	VARCHAR	255	Applicant's IP address and address combined. (PK)	901201-10-5678, 123 Jalan Utama
startDate	DATE		Application start date.	2023-10-26
endDate	DATE		Application end date.	2023-11-15

13. LoanCalc_History

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
historyID	INT	11	Unique identifier for each loan calculation history record. (PK)	701
memberID	VARCHAR	20	Foreign key referencing the Member table, indicating the member associated with this historical calculation. (FK)	M20230001
loanAmount	DECIMAL	11	The total amount of the loan	5000
loanDuration	INT	4	Duration of the loan (in months or years) in the historical calculation.	24
loanType	VARCHAR	50	Type of loan (e.g., Al-Bai, Al-Innah, etc.) in the historical calculation.	Al-Innah
interestRate	DECIMAL	5	Interest rate used in the historical calculation (e.g., 5.25 for 5.25%).	7.5

14.Record

FIELD NAME	DATA TYPE	FIELD SIZE	DESCRIPTION	EXAMPLE
record_id	INT	11	Unique identifier for each installment record. (PK)	500

4. Appendices

4.1 Use Case used for the system

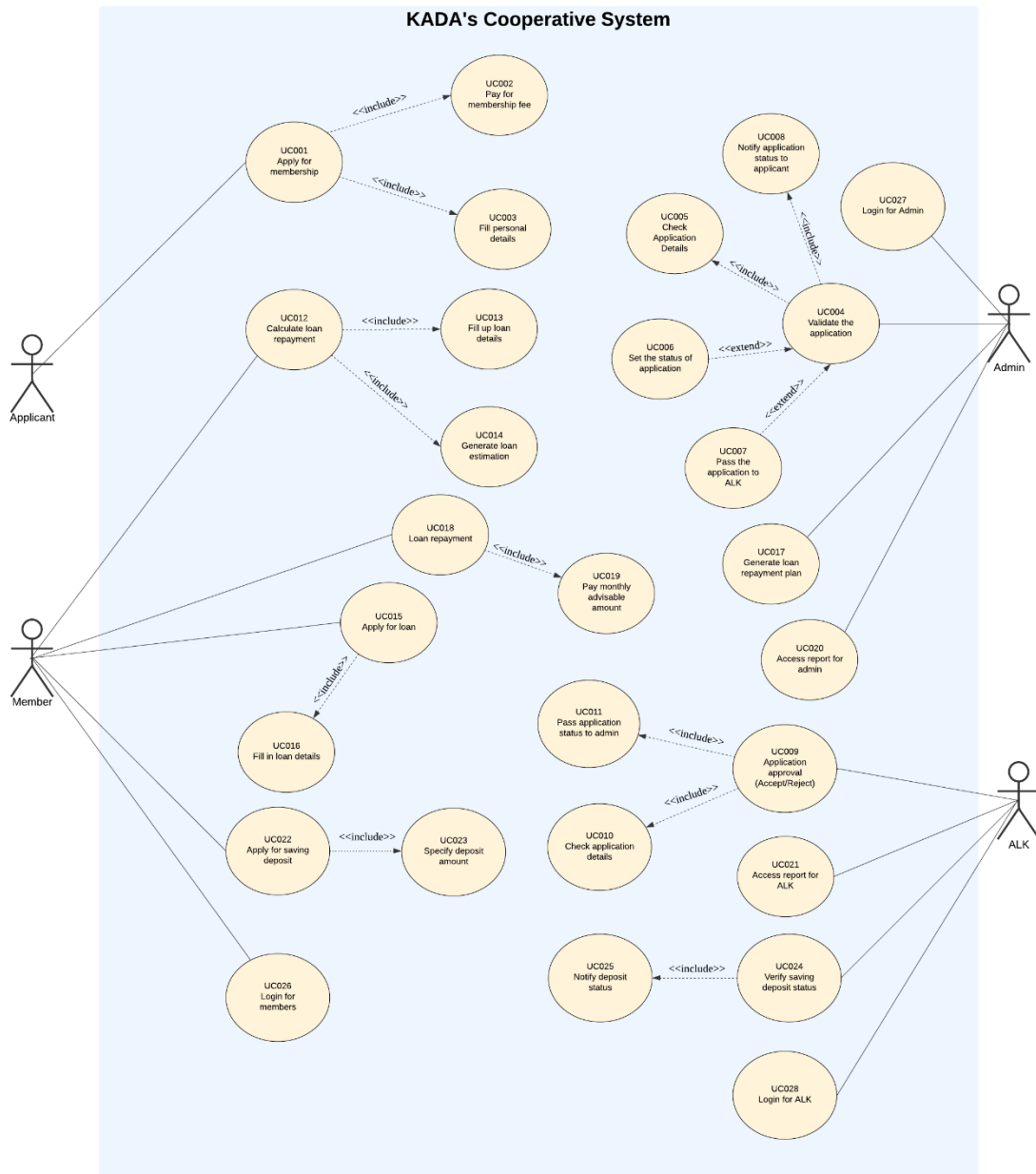
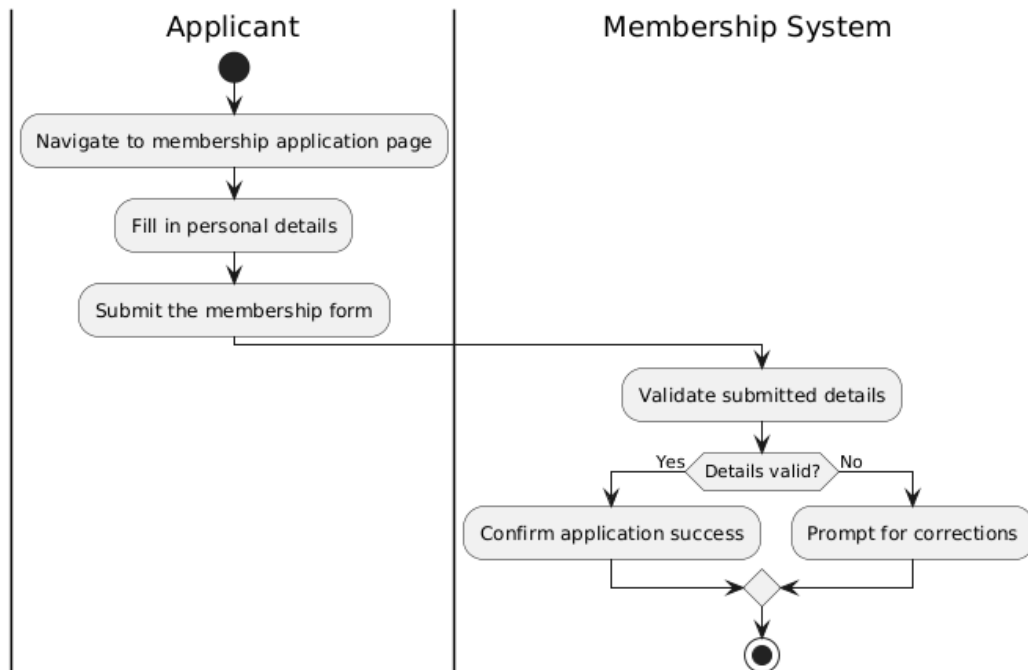


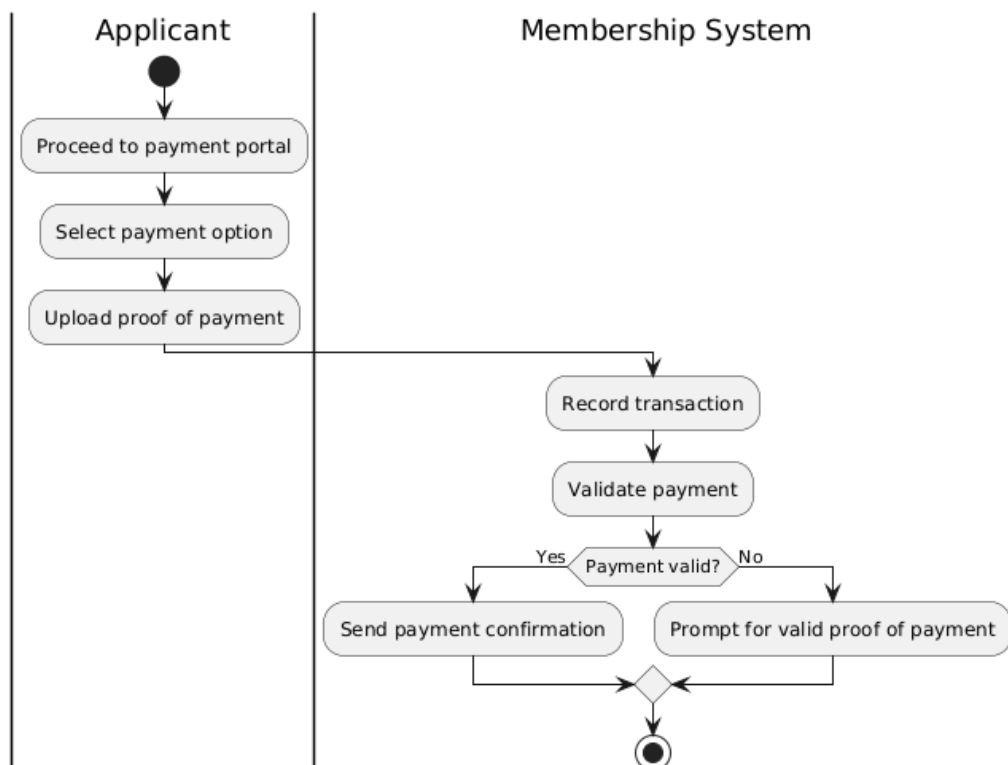
Figure 1.1 Usecase diagram for KADA's Cooperative System

4.2 Activity Diagram for Every Use Case Identified

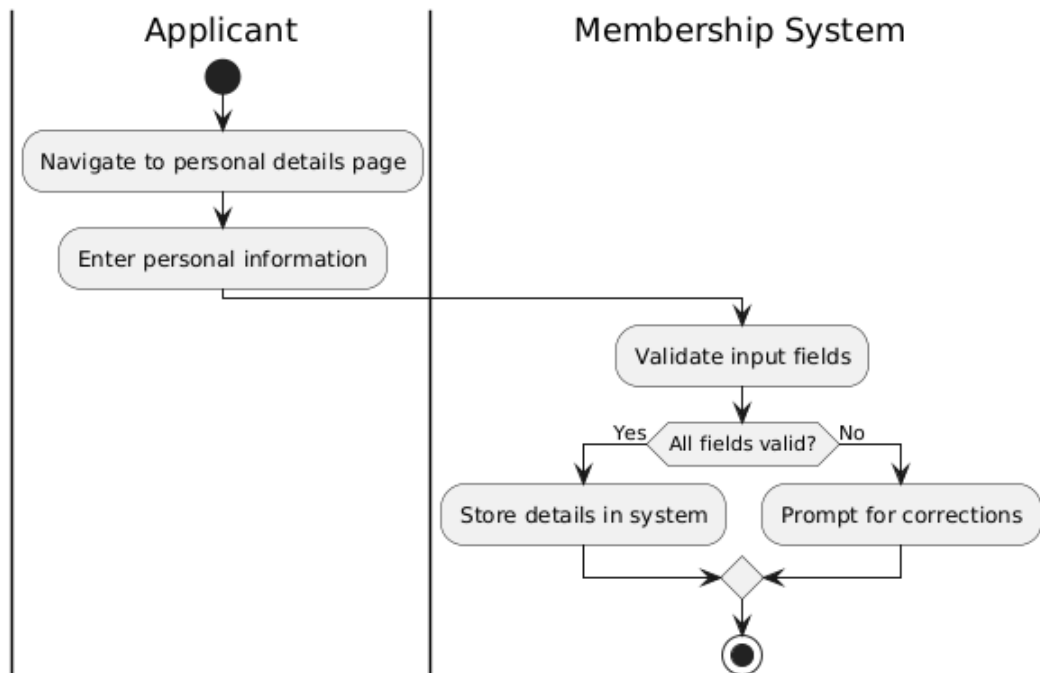
4.2.1 Activity Diagram 1: UC001



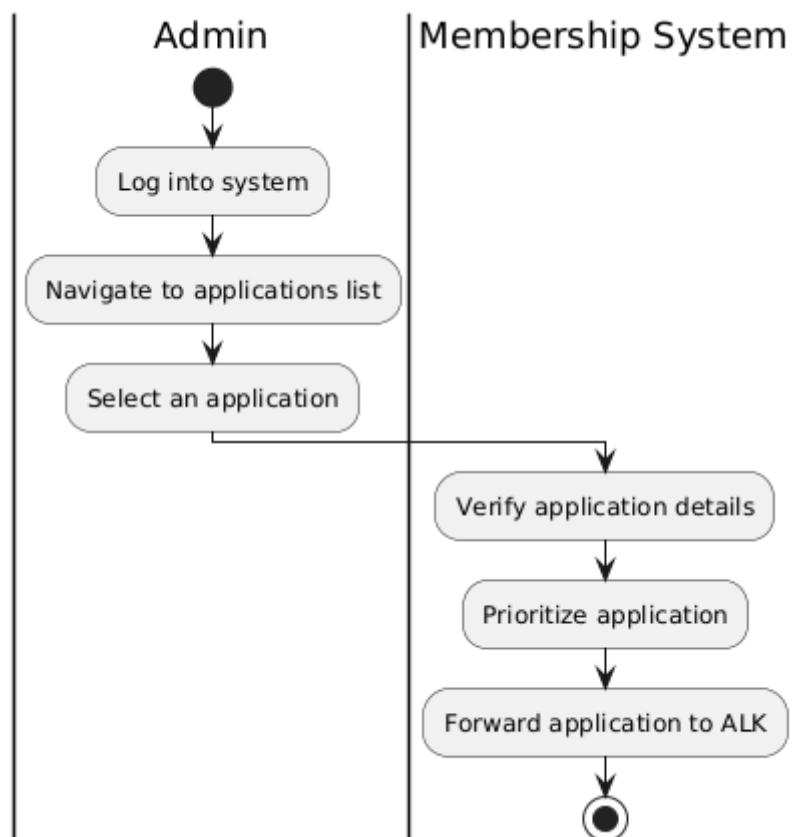
4.2.2 Activity Diagram 2: UC002



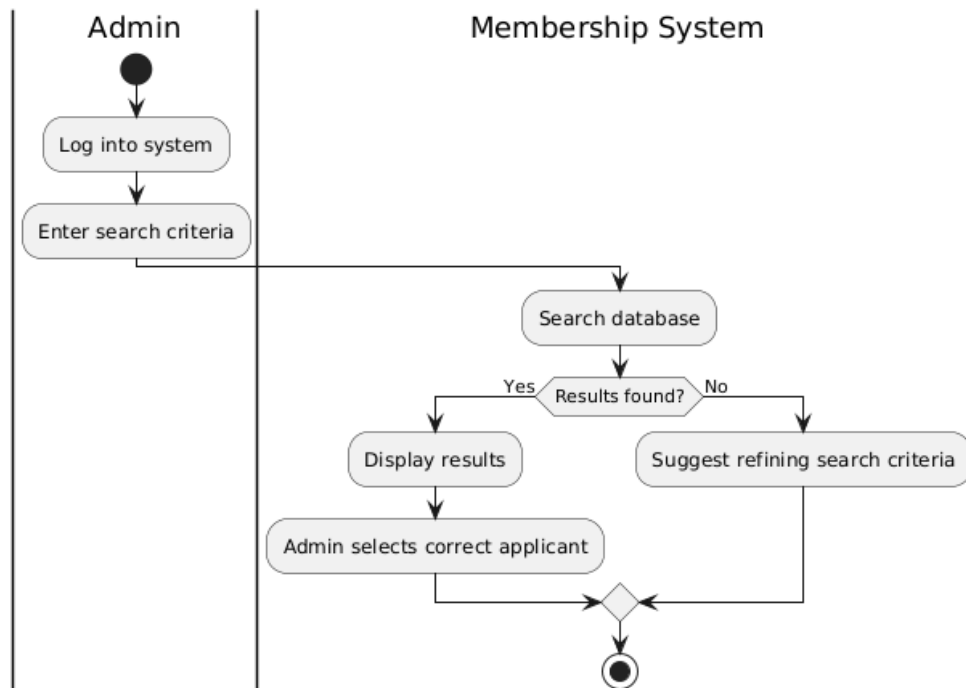
4.2.3 Activity Diagram 3: UC003



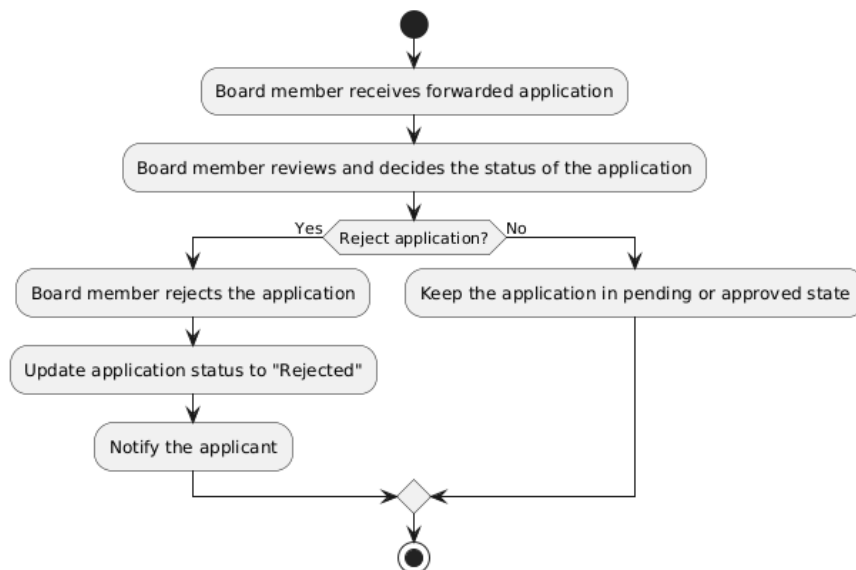
4.2.4 Activity Diagram 4: UC004



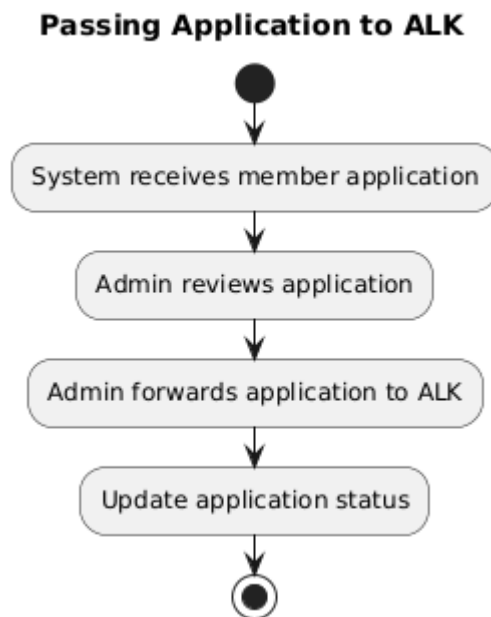
4.2.5 Activity Diagram 5: UC005



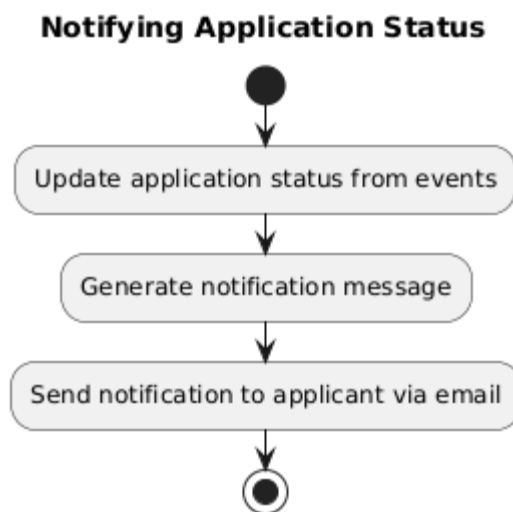
4.2.6 Activity Diagram 6: UC006



4.2.7 Activity Diagram 7: UC007

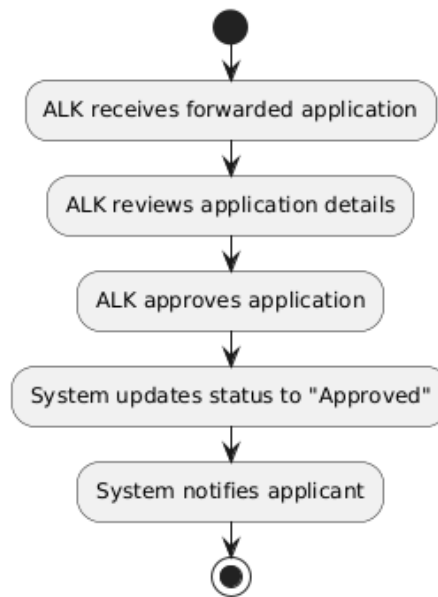


4.2.8 Activity Diagram 8: UC008

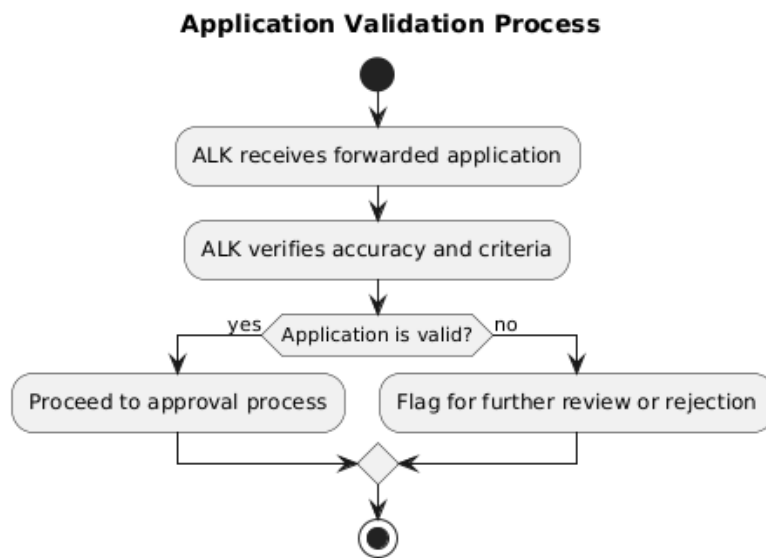


4.2.9 Activity Diagram 9: UC009

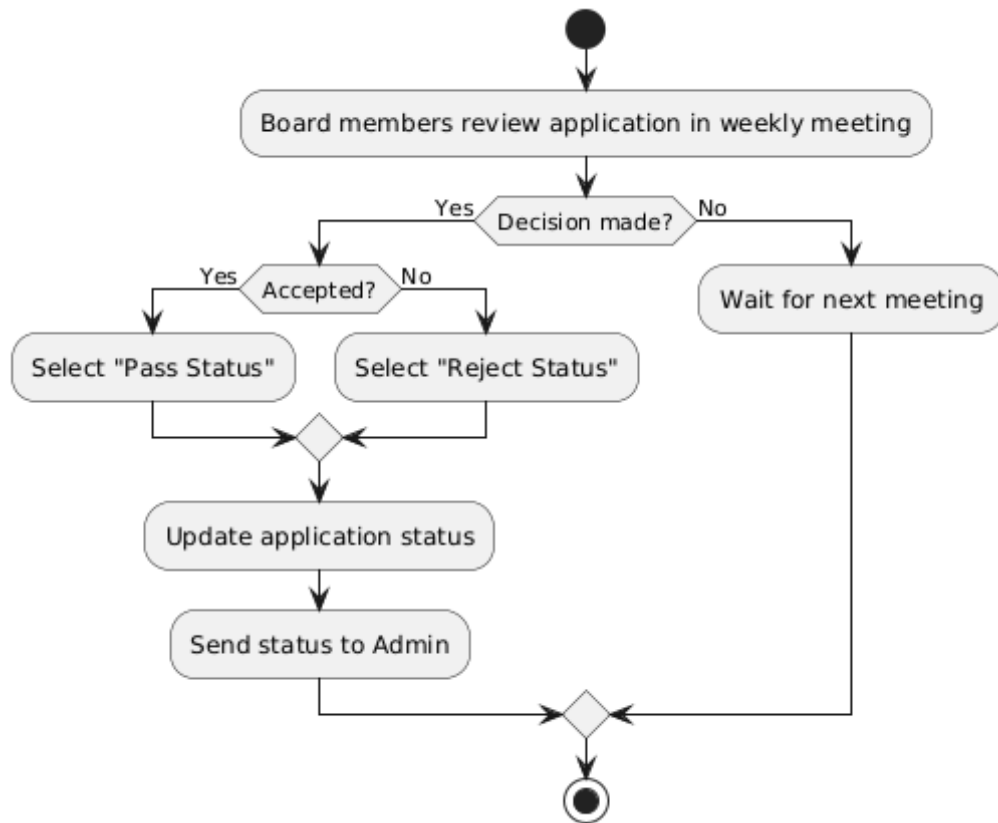
Application Approval Process



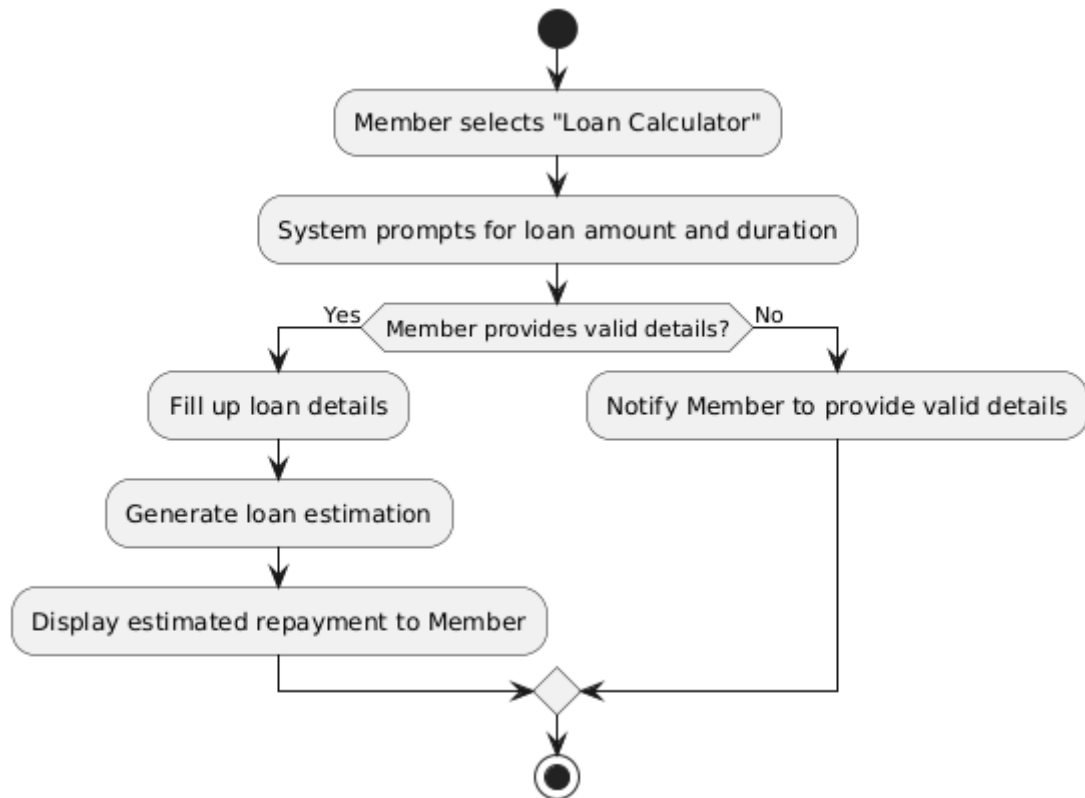
4.2.10 Activity Diagram 10: UC0010



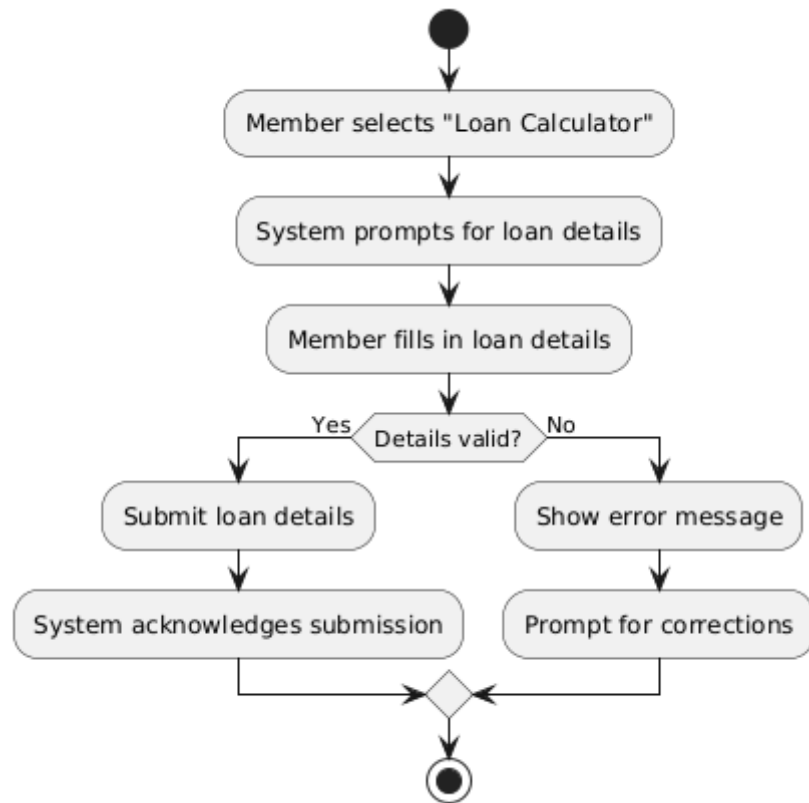
4.2.11 Activity Diagram 11: UC0011



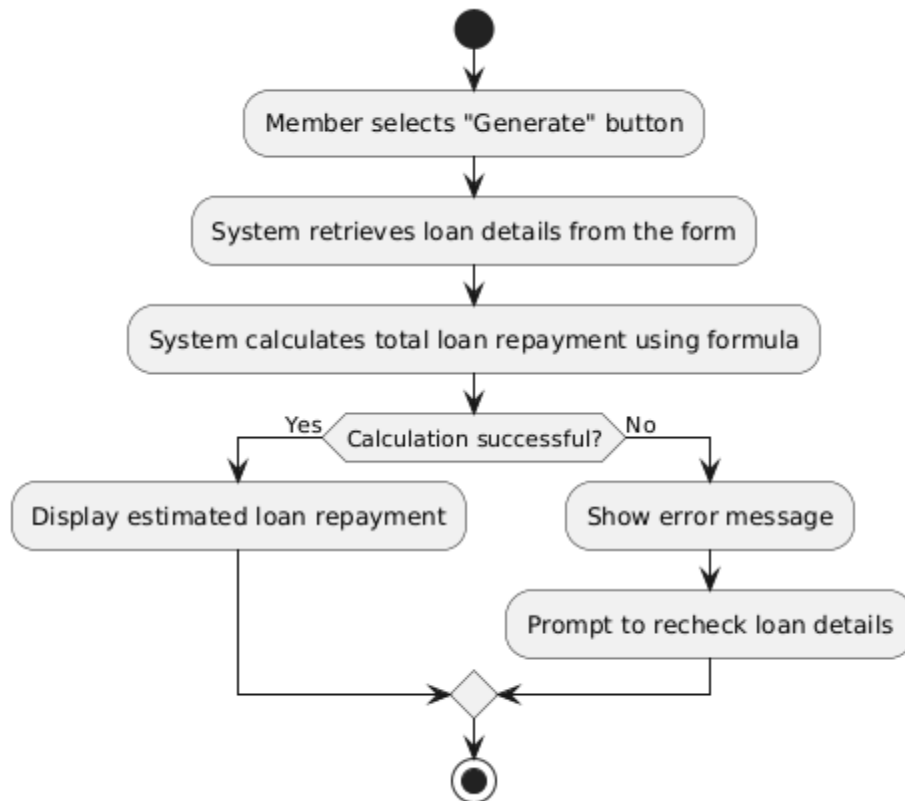
4.2.12 Activity Diagram 12: UC0012



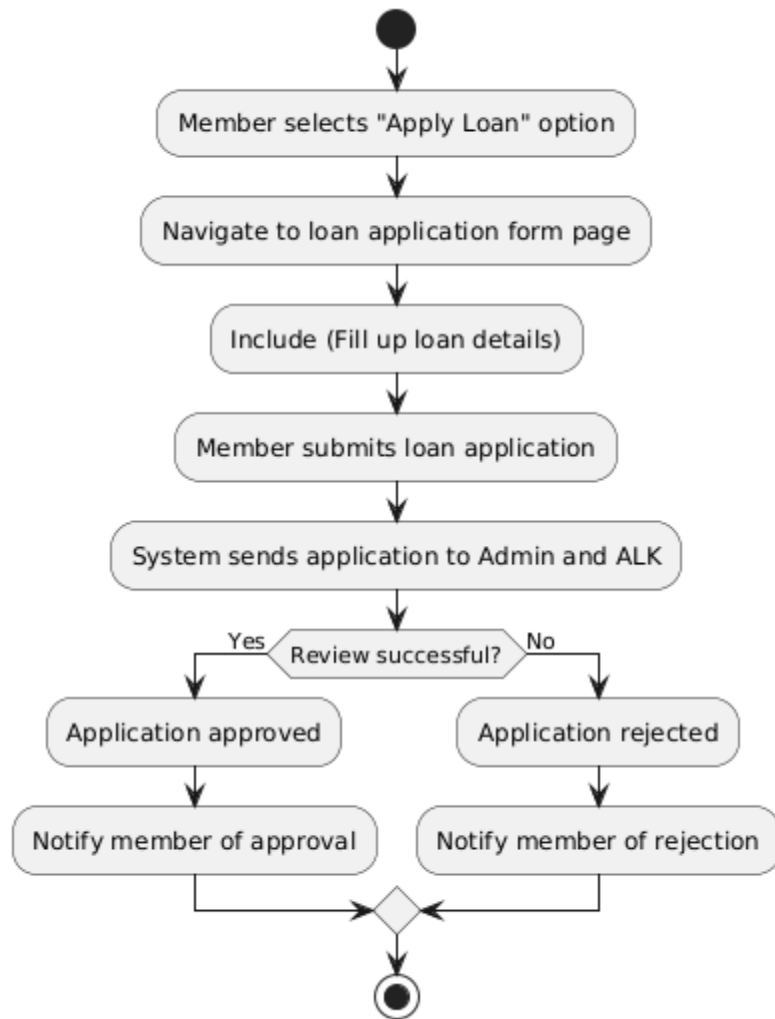
4.2.13 Activity Diagram 13: UC0013



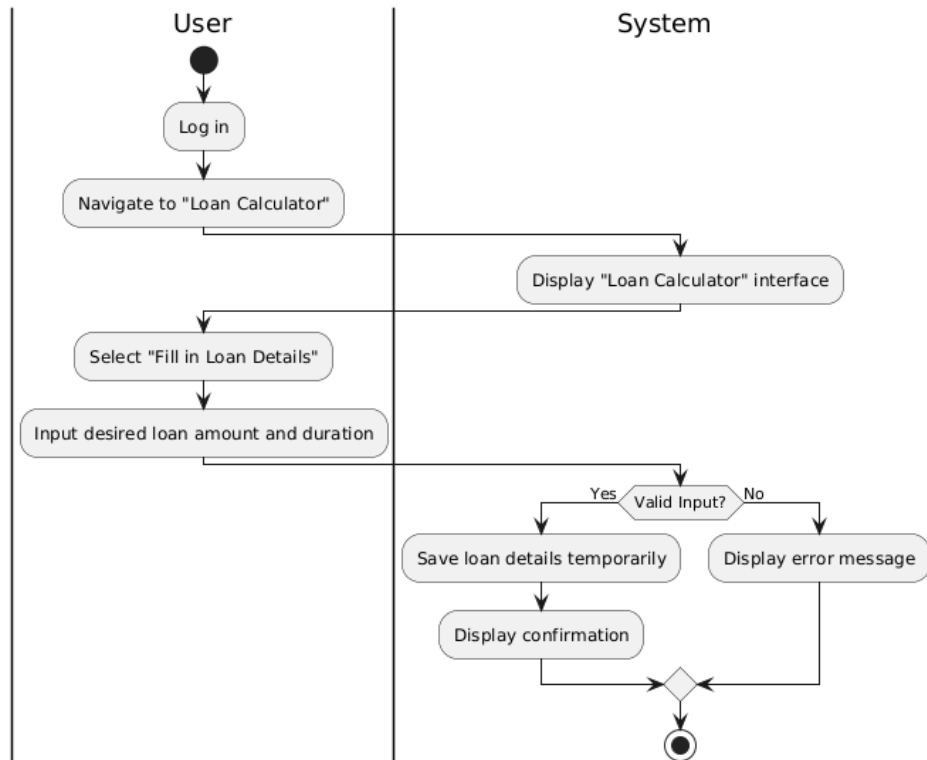
4.2.14 Activity Diagram 14: UC0014



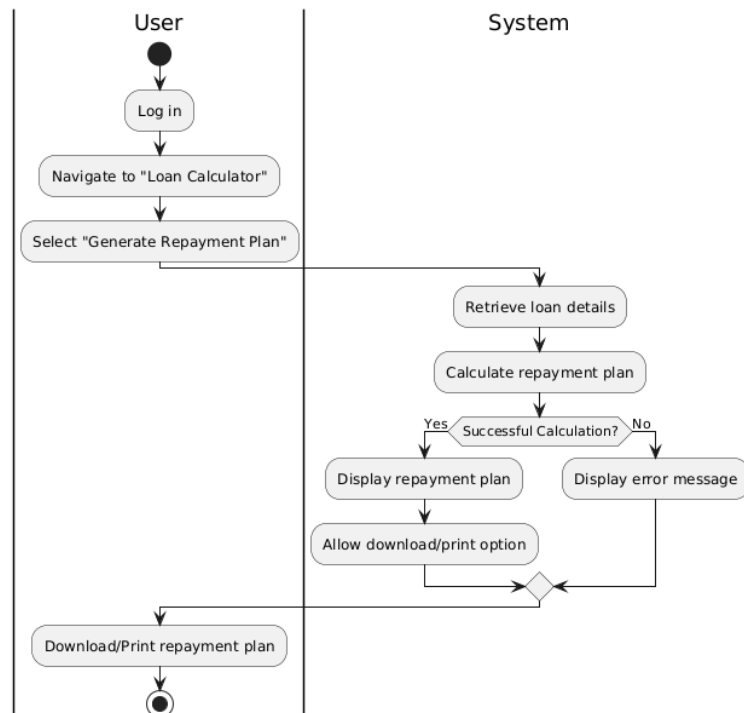
4.2.15 Activity Diagram 15: UC0015



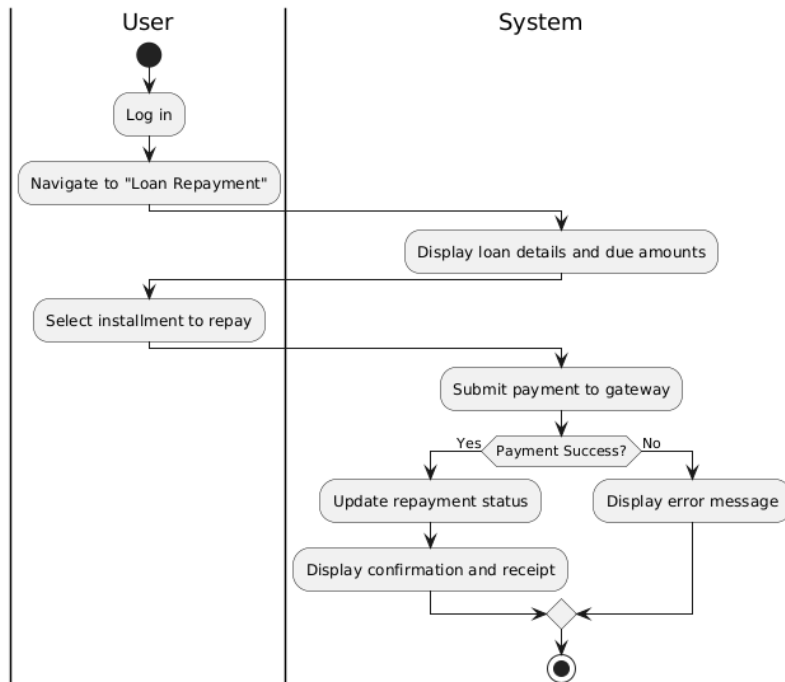
4.2.16 Activity Diagram 16: UC0016



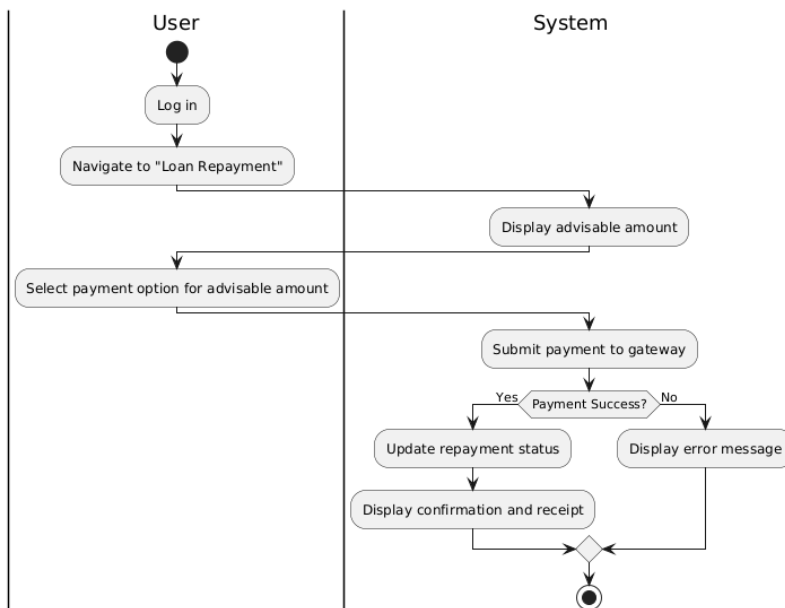
4.2.17 Activity Diagram 17: UC0017



4.2.18 Activity Diagram 18: UC0018



4.2.19 Activity Diagram 19: UC0019



4.3 Meeting Log (Google Meet)

DATE/TIME	29/11/2025 9.00 AM		
LOCATION	Google Meet		
AGENDA	i) Discuss about the task distribution ii) Discuss the detail about every task in P2		
Meeting MC	Muhammad Syahmi Faris Bin Rusli		
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Muhammad Afiq Danish bin Mohd Hazni		0900	
Muhammad Syahmi Faris bin Rusli		0900	
Muhammad Adam bin Razali		0900	
Ahmad Ziyaad bin Abbas		0900	
MINUTES			
NO.	ITEMS DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Task Distribution	All group members have their own task that they need to do and complete	Syahmi Afiq Adam Ziyaad
2	Discuss the details	All group members give their idea and opinion on what they understand about the P2	Syahmi Afiq Adam Ziyaad

